

Ishikawa (Fishbone) Analysis for Security Incident Mapping (SIM)

Problem Statement

High Rate of Campus Insecurity

The major causes of campus insecurity can be grouped into the following categories:

1. People

- Students engaging in criminal activities
- Drug and substance abuse
- Cultism and gang violence
- Lack of security awareness
- Fear of reporting crimes
- Low student cooperation with security personnel

2. Process

- Slow reporting of incidents
- No centralized reporting system
- Poor emergency response procedures
- Manual record keeping
- Poor communication between victims and security officers

3. Environment

- Poor street lighting
- Isolated pathways
- Unsecured hostels and residences
- Overcrowded student communities
- Bushy or abandoned areas around campus

4. Technology

- Lack of a digital crime reporting platform
- No real-time notifications
- Poor internet connectivity in some locations
- Absence of incident mapping tools
- Limited surveillance equipment

5. Management

- Inadequate campus security personnel

- Poor monitoring and supervision
- Insufficient funding for security infrastructure
- Weak enforcement of security policies
- Delayed response to reported incidents

6. Equipment

- Insufficient CCTV cameras
- Faulty communication devices
- Lack of patrol vehicles
- Poor lighting equipment
- Limited emergency response tools

Proposed Solution

The Security Incident Mapping (SIM) web application addresses these issues by providing:

- A centralized online platform for reporting incidents.
- Faster communication between users and security personnel.
- Digital storage of incident records.
- Better monitoring of crime-prone locations.
- Easier access to emergency contact information.
- Improved decision-making through organized incident data.

Conclusion

The Ishikawa analysis shows that campus insecurity is caused by multiple interconnected factors involving people, processes, technology, management, the environment, and equipment. Implementing the SIM system can help reduce these challenges by improving reporting, communication, and incident management.